

A Critical Replication Under Data Corruption and New Offline RL Settings

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Introduction

Paper: Xu et al., Tackling Data Corruption in Offline Reinforcement Learning via Sequence Modeling – ICLR 2025.

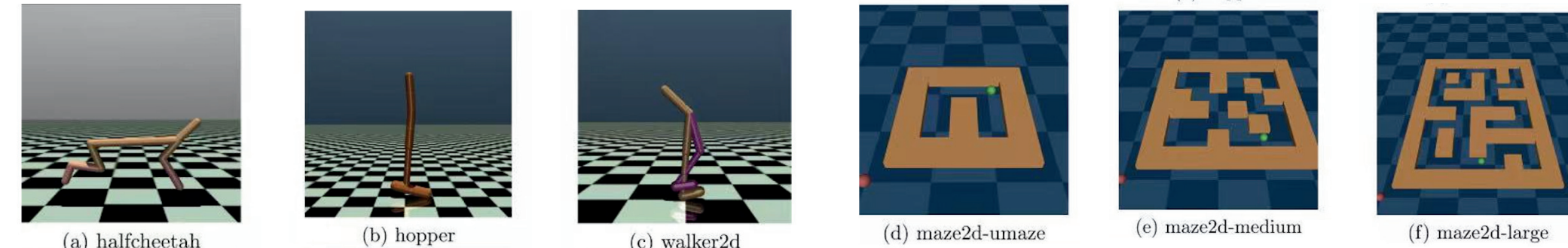


Figure 1. Offline RL benchmark tasks used to evaluate robustness, including locomotion environments and maze navigation tasks.

Real-world offline RL data is often noisy, mis-labeled, or partially adversarial, which can make learned policies unreliable. **Robust Decision Transformer** argues that **sequence-modeling** methods are more robust to corrupted offline data than **TD-based offline RL** methods, and proposes RDT by adding three robustness modules to Decision Transformer: embedding dropout, Gaussian-weighted learning, and iterative data correction. In this project, we **critically replicate** these claims by testing whether DT outperforms RIQL under corruption, whether RDT consistently improves over DT, and where these robustness claims fail under harder settings.

Objective

Claim 1: DT is more robust than TD-based methods under corrupted offline RL data.

Claim 2: RDT improves over vanilla DT across different corruption types.

Claim 3: RDT remains robust under various severity, dataset and corruption conditions

Methodology

1 We conducted a critical replication of Robust Decision Transformer by comparing three offline RL methods: **RIQL**, **Decision Transformer**, and **RDT**.

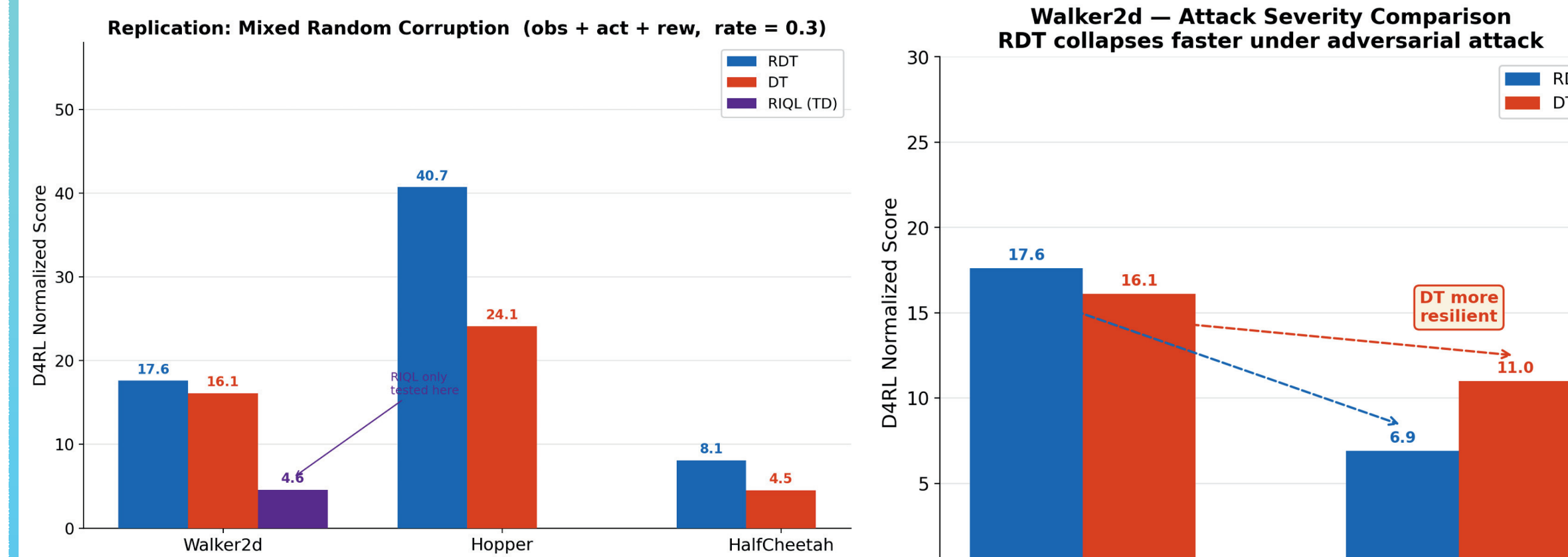
2 The original RDT paper evaluates robustness on three offline RL benchmark families: MuJoCo, Kitchen, and Adroit. In our replication, we focus on **MuJoCo locomotion tasks** – Walker2d, Hopper, and HalfCheetah – and add AntMaze as a new held-out sparse-reward navigation task to **test generalization** beyond the original setup.

3 We introduced different types of training-time data corruption, including observation/state corruption, action corruption, reward corruption, mixed-random corruption, and adversarial-mixed corruption.

4 We evaluated **robustness** using **normalized return scores**, comparing DT vs. RIQL to test sequence modeling against TD-learning, and RDT vs. DT to test whether RDT adds robustness.

5 We tested RDT's three modules through ablations: **embedding dropout**, **Gaussian-weighted MSE**, and **iterative data correction**, to see which components helped under different corruption types.

Replication of Paper



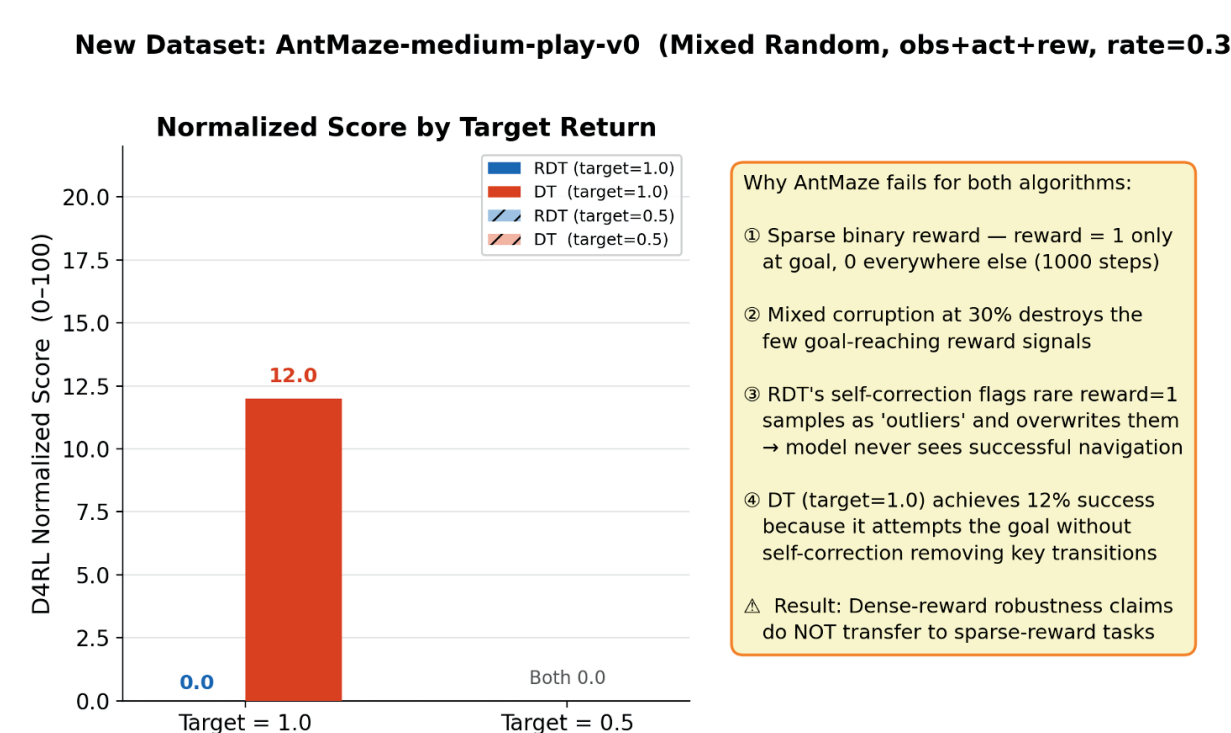
Claim Verification

Claim	Setting	RDT Score	DT Score	Verdict
DT > TD-learning	Walker2d, mixed random	16.1	4.6 (RIQL)	CONFIRMED
RDT > DT (action)	Walker2d, act-only random	31.6	27.7	CONFIRMED
RDT > DT (reward)	Walker2d, rew-only random	40.3	37.6	CONFIRMED
RDT > DT (state)	Hopper, obs-only random	27.6	31.9	FAILS
RDT > DT (mixed adv)	Walker2d, adv mixed	6.9	11.0	FAILS
RDT > DT (mixed)	Hopper, mixed random	40.7	24.1	CONFIRMED
Block > Random	Walker2d, mixed (RDT)	19.9	17.6 (rand)	NEW FINDING
New dataset	AntMaze, mixed random	0.0	12.0	NEW FINDING

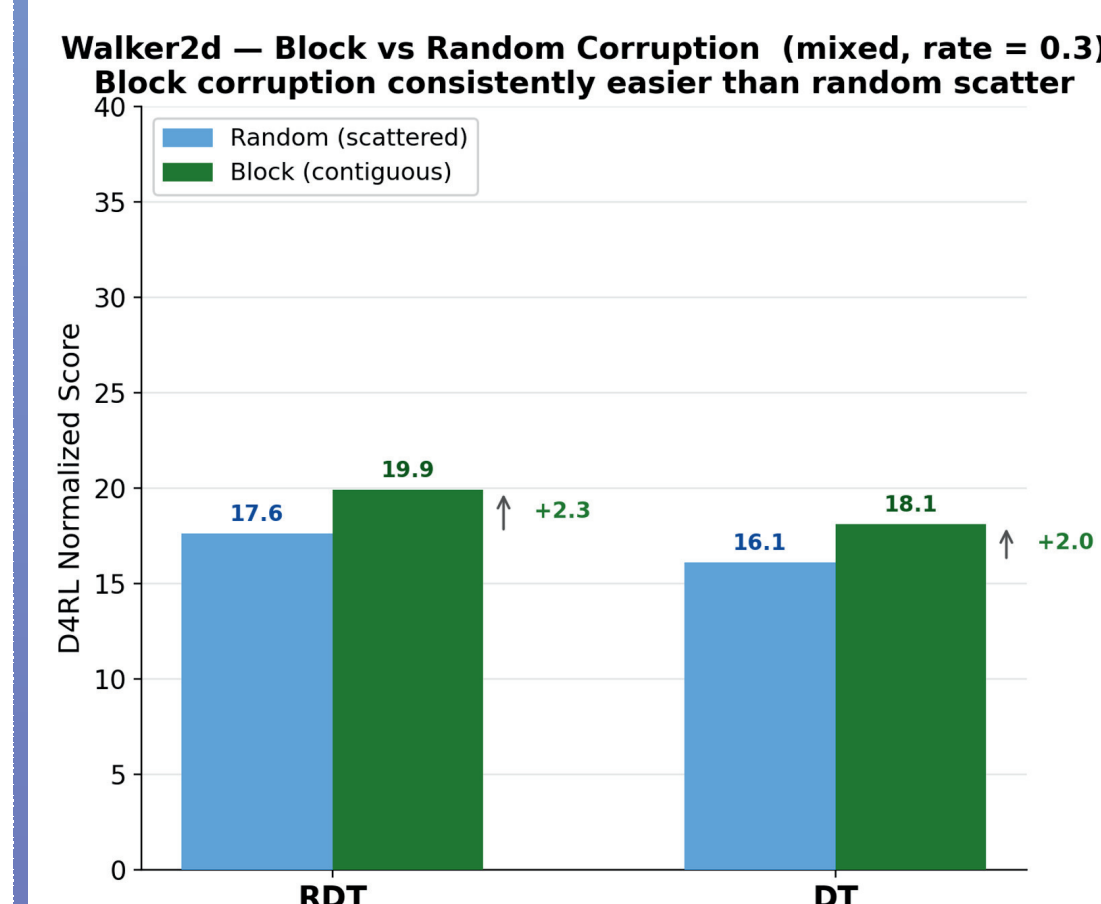
Hypothesis-Driven Ablations

Setting	Score(high)	Delta vs full RDT
Full RDT (state attack)	19.9	-
RDT - embedding dropout	19.0	-0.9
RDT - weighted MSE	24.3	+4.4
RDT - correction (action)	31.6	+3.6 vs RDT 28.0
DT (no robustness)	22.7	+2.8

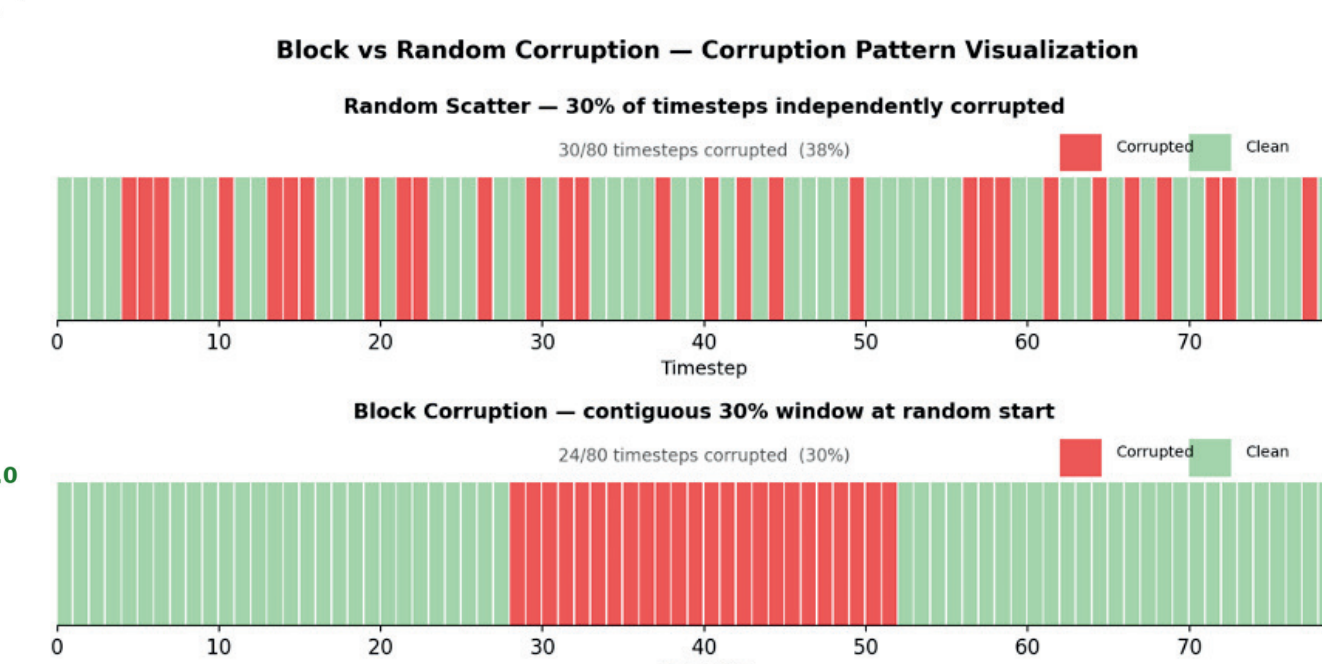
New dataset - AntMaze-medium-play-v0



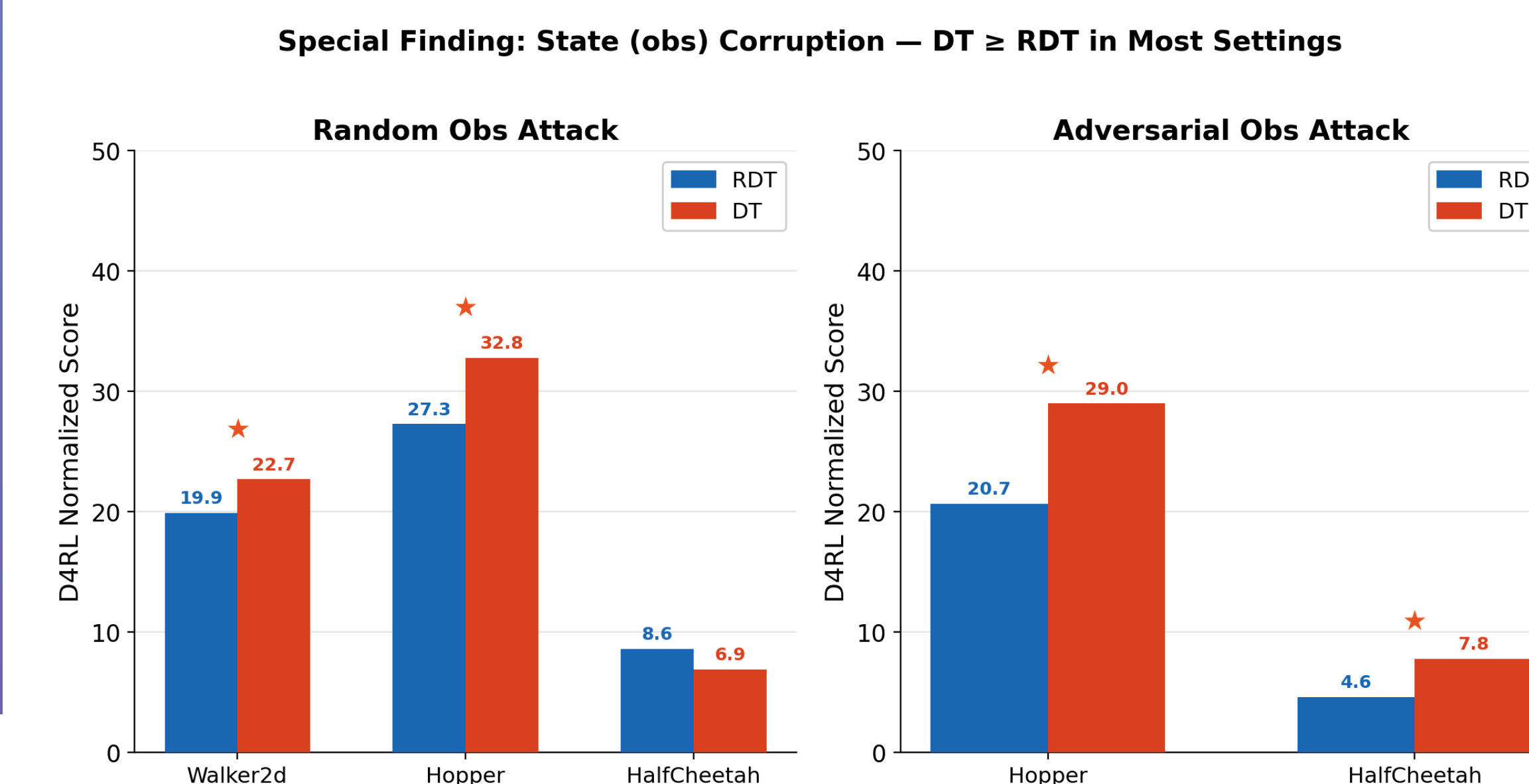
Block (contiguous) vs random corruption



Block (contiguous) corruption is easier



Under state attack, RDT < DT



Results

Our replication partially **confirms the RDT paper's claims**.

Claim 1 confirmed: DT outperformed the TD-based baseline RIQL under Walker2d mixed-random corruption: **DT 16.1 vs. RIQL 4.6**

Claim 2 partially confirmed: RDT improved over DT in **4/6 corruption settings**, especially under action, reward, and mixed-random corruption. RDT failed under **Hopper obs-only random** and **Walker2d adversarial mixed** corruption.

Claim 3 not fully confirmed: RDT did not generalize reliably to harder stress tests. It handled block corruption slightly better than scattered random corruption, but failed on the sparse-reward AntMaze task: **RDT 0.0 vs. DT 12.0**

Key takeaway: RDT is usually more robust than DT, but its advantage is **not universal**. Under harder adversarial, observation-only, or sparse-reward settings, its robustness can break down.

Discussion

Lessons learned:

Our results suggest that RDT's robustness is **conditional** rather than **universal**. RDT improves over DT in several corrupted-data settings, especially **action, reward, and mixed-random corruption**, but it can fail under state/observation corruption, mixed adversarial corruption, and sparse-reward AntMaze. This shows that the three RDT modules do not always stack additively; sometimes a robustness mechanism can misfire when prediction error comes from **noisy inputs** instead of corrupted labels.

Limitations:

Our implementation was limited by **compute, environment compatibility, and reproducibility challenges**. We used a newer MuJoCo setup than the original paper, and full sweeps over all environments, attacks, and random seeds were too expensive. Because of this, our results should be interpreted as a targeted replication rather than a full reproduction of every original experiment.

Future work:

Future work could **run more seeds, test more datasets, and separate the effects** of each corruption type more carefully. A useful next step would be redesigning RDT's correction mechanism for sparse-reward tasks, so that rare successful transitions are preserved instead of being incorrectly treated as outliers.

